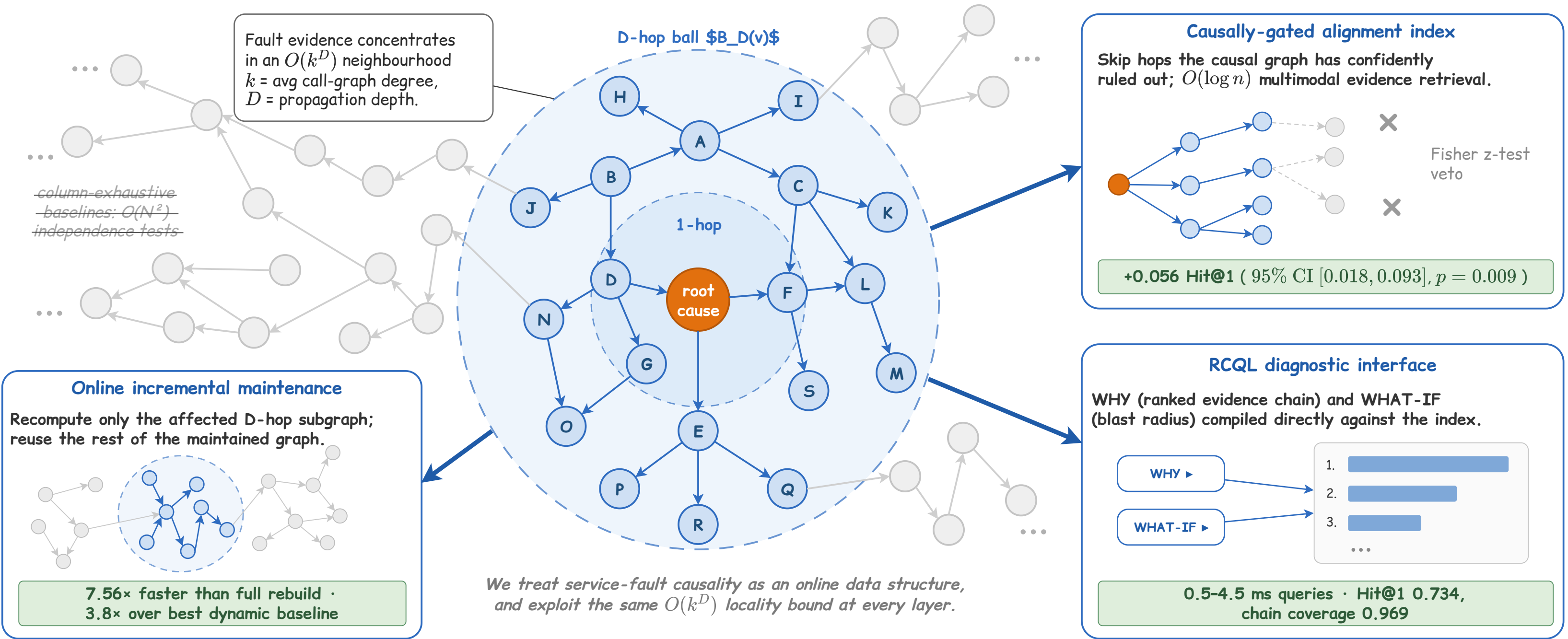




Fault evidence concentrates in an $O(k^D)$ neighbourhood
 k = avg call-graph degree,
 D = propagation depth.

~~column-exhaustive~~
~~baselines: $O(N^2)$~~
~~independence tests~~

D-hop ball $B_D(v)$

1-hop

root cause

We treat service-fault causality as an online data structure,
 and exploit the same $O(k^D)$ locality bound at every layer.

Fisher z-test veto

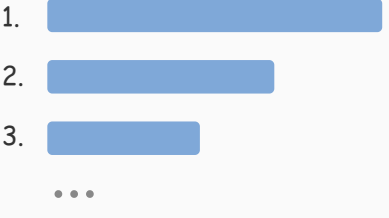
+0.056 Hit@1 (95% CI [0.018, 0.093], $p = 0.009$)

RCQL diagnostic interface

WHY (ranked evidence chain) and WHAT-IF (blast radius) compiled directly against the index.

WHY ▶

WHAT-IF ▶



0.5-4.5 ms queries · Hit@1 0.734,
 chain coverage 0.969